

Question-1:

Create 1 Public Docker Hub registry named flask_webapp_yourname.

- Clone below repository on your system.

<https://github.com/mmumshad/simple-webapp-docker.git>

- Initialize a local repository & copy the code from above repo to your local repository in your working branch.

- Once code is copied to the local repository, build the docker image & add meaningful tags with version 1 and push to Docker Hub registry.

- Once Images are pushed to Docker hub registries, create a directory named kube.

Inside the kube directory create deployment.yaml file with 3 replication , labels app:

flask-webapp , containerPort: 8080 and add the image that you have pushed in Docker

Hub registry.

- Create one service.yaml file with type nodeport & select flask-webapp with port 8080 &

targetPort 8080 with any nodePort between range 30000-32768.

- Once a service is created try accessing the web page in the browser as

below. (30011 is

nodeport mentioned in service.yaml). Meanwhile open app.py from your code to understand paths & output.

http://master_ip:30011/

[http://master_ip:30011/how are you](http://master_ip:30011/how%20are%20you)

- Now , update the app.py from your code and add below route above if `__name__ ==`

`"__main__"` line

```
@app.route('/Who are you')
```

```
def cloudeithix():
```

```
    return 'Yes, I am cloudeithix, and You !!!'
```

- Once the file is updated , rebuild the docker image & add meaningful tags with version 2

and push to Docker Hub registry.

- Now we have the latest docker image in repo, It's time to roll out a new image. Roll out

the new Image with all three ways one by one

With kubectl set command

2. With kubectl edit deployment

3. With deployment.yaml file modification.

- Run the `# kubectl rollout` command to check status and history.

- Note:- Once above step 1 is done , run `# kubectl rollout undo deployment` command to

rollback the change and then try a second way of rollout.

- In the browser run all three routes & notice the changes.

http://master_ip:30011/

[http://master_ip:30011/how are you](http://master_ip:30011/how%20are%20you)

[http://master_ip:30011/Who are you](http://master_ip:30011/Who%20are%20you)

- Once done with all above steps , commit all the changes to the remote repository.

- Capture the snap and prepare a well formatted document.

```
=====
payal_kharat@LAPTOP-# git clone https://github.com/mmumshad/simple-webapp-docker.git
```

```
Cloning into 'simple-webapp-docker'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 14 (delta 3), reused 2 (delta 2), pack-reused 7 (from 1)
Receiving objects: 100% (14/14), done.
Resolving deltas: 100% (3/3), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-1$ ls
simple-webapp-docker

-----

payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/Assignment-3/Que-1/.git/
payal_kharat@LAPTOP-# cp -rf simple-webapp-docker/* .
payal_kharat@LAPTOP-# ls
Dockerfile app.py simple-webapp-docker

-----

payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: flask-webapp-revision1
  namespace: cloudethix-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: flask-webapp-revision1
  template:
    metadata:
      labels:
        app: flask-webapp-revision1
    spec:
      containers:
        - name: flask-webapp-revision1
          image: docker.io/payalkharat/flask_webapp_payal:v1
          ports:
            - containerPort: 8080

-----

payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/flask-webapp-revision1 created
payal_kharat@LAPTOP-# kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE
IP                                  NODE    NOMINATED NODE   READINESS GATES
```

```
flask-webapp-revision1-679449b7bf-krfwj    1/1      Running    0          5m43s
10.244.2.2    mycluster-worker    <none>    <none>
flask-webapp-revision1-679449b7bf-mnddb    1/1      Running    0          5m43s
10.244.1.3    mycluster-worker2    <none>    <none>
flask-webapp-revision1-679449b7bf-rszd5    1/1      Running    0          5m43s
10.244.1.2    mycluster-worker2    <none>    <none>
payal_kharat@LAPTOP-V1NV0QFT:Que-1$ kubectl get deployment
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
flask-webapp-revision1             3/3      3              3            5m53s
```

```
-----
payal_kharat@LAPTOP-# cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: flask-webapp-service
  namespace: cloudethix-1
spec:
  type: NodePort
  selector:
    app: flask-webapp-revision1
  ports:
    - port: 8080
      targetPort: 8080
      nodePort: 30012
```

```
-----
payal_kharat@LAPTOP-# kubectl apply -f service.yaml
service/flask-webapp-service created
payal_kharat@LAPTOP-# kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)
AGE
flask-webapp-service               NodePort       10.96.114.241   <none>           8080:30012/TCP
5s
```

```
-----
payal_kharat@LAPTOP-# cat app.py
import os
from flask import Flask
app = Flask(__name__)

@app.route("/")
def main():
    return "Welcome!"

@app.route('/how are you')
def hello():
    return 'I am good, how about you?'

@app.route('/Who are you')
def cloudethix():
    return 'Yes, I am cloudethix, and You !!!'

if __name__ == "__main__":
    app.run()
```

```
-----
payal_kharat@LAPTOP-# kubectl set image deployment/flask-webapp-revision1
flask-webapp-revision1=docker.io/payalkharat/flask_webapp_payal:v2
deployment.apps/flask-webapp-revision1 image updated
payal_kharat@LAPTOP-# kubectl rollout status deployment/flask-webapp-revision1
deployment "flask-webapp-revision1" successfully rolled out
-----
payal_kharat@LAPTOP-# kubectl get deployment
```

```
NAME                READY    UP-TO-DATE    AVAILABLE    AGE
flask-webapp-revision1  3/3      3              3             13m

payal_kharat@LAPTOP-# kubectl get pods
NAME                READY    STATUS    RESTARTS    AGE
flask-webapp-revision1-679449b7bf-6ndk9  1/1      Running    0            8s
flask-webapp-revision1-679449b7bf-pssr9  1/1      Running    0            5s
flask-webapp-revision1-679449b7bf-v4zcz  1/1      Running    0            7s
flask-webapp-revision1-f99d5d98-5mgjx    1/1      Terminating  0
3m31s
flask-webapp-revision1-f99d5d98-98fs7    1/1      Terminating  0
3m47s
flask-webapp-revision1-f99d5d98-ssg6v    1/1      Terminating  0
3m38s

-----

payal_kharat@LAPTOP-# kubectl rollout undo deployment/flask-webapp-revision1
deployment.apps/flask-webapp-revision1 rolled back

-----

payal_kharat@LAPTOP-# vim deployment.yaml
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: flask-webapp-revision1
  namespace: cloudeithix-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: flask-webapp-revision1
  template:
    metadata:
      labels:
        app: flask-webapp-revision1
    spec:
      containers:
        - name: flask-webapp-revision1
          image: docker.io/payalkharat/flask_webapp_payal:v2
          ports:
            - containerPort: 8080

-----

payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/flask-webapp-revision1 configured
payal_kharat@LAPTOP-# kubectl rollout status deployment/flask-webapp-revision1
deployment "flask-webapp-revision1" successfully rolled out

-----

payal_kharat@LAPTOP-# kubectl rollout history deployment/flask-webapp-
revision1
deployment.apps/flask-webapp-revision1
REVISION    CHANGE-CAUSE
3            <none>
4            <none>

-----

payal_kharat@LAPTOP-# kubectl get pods
NAME                READY    STATUS    RESTARTS    AGE
flask-webapp-revision1-679449b7bf-6ndk9  1/1      Terminating  0
2m52s
flask-webapp-revision1-679449b7bf-pssr9  1/1      Terminating  0
2m49s
flask-webapp-revision1-679449b7bf-v4zcz  1/1      Terminating  0
2m51s
```

flask-webapp-revision1-f99d5d98-7nwxn	1/1	Running	0	28s
flask-webapp-revision1-f99d5d98-n8knd	1/1	Running	0	26s
flask-webapp-revision1-f99d5d98-nz76v	1/1	Running	0	30s

payal_kharat@LAPTOP-# kubectl rollout undo deployment/flask-webapp-revision1
deployment.apps/flask-webapp-revision1 rolled back

payal_kharat@LAPTOP-# kubectl edit deployment flask-webapp-revision1
deployment.apps/flask-webapp-revision1 edited
payal_kharat@LAPTOP-# kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
flask-webapp-revision1-679449b7bf-644pt	1/1	Terminating	0	100s
flask-webapp-revision1-679449b7bf-zlxh5	1/1	Terminating	0	101s
flask-webapp-revision1-f99d5d98-bg7hl	1/1	Running	0	31s
flask-webapp-revision1-f99d5d98-kq9jc	1/1	Running	0	32s
flask-webapp-revision1-f99d5d98-lm7mp	1/1	Running	0	29s

payal_kharat@LAPTOP-# kubectl rollout status deployment/flask-webapp-revision1
deployment "flask-webapp-revision1" successfully rolled out
payal_kharat@LAPTOP-# kubectl rollout history deployment/flask-webapp-revision1
deployment.apps/flask-webapp-revision1

REVISION	CHANGE-CAUSE
5	<none>
6	<none>

payal_kharat@LAPTOP-# curl http://localhost:8080
Welcome!
payal_kharat@LAPTOP-# curl http://localhost:8080/how%20are%20you
I am good, how about you?
payal_kharat@LAPTOP-# curl http://localhost:8080/Who%20are%20you
Yes, I am cloudethix, and You !!!

payal_kharat@LAPTOP-V1NV0QFT:Que-1\$ git status
On branch master

No commits yet

Untracked files:
 (use "git add <file>..." to include in what will be committed)
 Dockerfile
 app.py
 deployment.yaml
 service.yaml

nothing added to commit but untracked files present (use "git add" to track)
payal_kharat@LAPTOP-# git add .
payal_kharat@LAPTOP-# git commit -m "second revision updated webapp"
[master (root-commit) 5a79dba] second revision updated webapp
 4 files changed, 61 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 deployment.yaml
 create mode 100644 service.yaml

payal_kharat@LAPTOP-# git push -u origin master
Username for 'https://github.com': CloudEthix987
Password for 'https://CloudEthix987@github.com':

```
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.04 KiB | 35.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/CloudEthix987/-
flask_webapp_payal/pull/new/master
remote:
To https://github.com/CloudEthix987/-flask_webapp_payal.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

=====
```

Question-3

- Clone below repository on your system.
`https://github.com/zembutsu/docker-sample-nginx.git`
- Initialize a local repository & copy the code from above repo to your local repository in the working branch.
- Once code is copied , build a docker image from docker file and add meaningful tags and push to docker hub repository
Once Images are pushed to Docker hub registries, create a directory named kube.
Inside the kube directory create deployment.yaml file with 3 replication , labels app: frontend-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.
- Create one service.yaml file with type nodeport & select frontend-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
- Once the service is created try accessing the web page in the browser as below. Notice the changes & take the snap.
- Now create a configmap.yaml file with below data & delete the deployment that you have created.
<html>
<body>
<h1> I am Cloudethix Team, Are you ?!! </h1>
Version: 1.1
</body>
</html>
- Then update the same deployment.yaml file and mount configmap as volume on container using volumeMounts with mountPath /usr/share/nginx/html/
- Now it's time to create configmap & deployment. Once created , try to access the webpage in the browser & confirm that the index page is the same as we have in configmap.

System Output
=====

```
payal_kharat@LAPTOP-# git clone https://github.com/zembutsu/docker-sample-nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-3$ ls
docker-sample-nginx
payal_kharat@LAPTOP-V1NV0QFT:Que-3$ cp -rf docker-sample-nginx/* .
```

```
payal_kharat@LAPTOP-# ls
Dockerfile LICENSE README.md default.conf index.html
payal_kharat@LAPTOP-V1NV0QFT:Que-3$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/Assignment-3/Que-3/.git/
```

```
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-webapp-revision1
  namespace: cloudethix-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend-webapp-revision1
  template:
    metadata:
      labels:
        app: frontend-webapp-revision1
    spec:
      containers:
        - name: frontend-webapp-revision1
          image: docker.io/payalkharat/cloudethix_configmap_payal:1.0
          ports:
            - containerPort: 80
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/frontend-webapp-revision1 created
```

```
payal_kharat@LAPTOP-# kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
frontend-webapp-revision1-65c49c4776-55csm  1/1     Running   0           100s
frontend-webapp-revision1-65c49c4776-gcwpd   1/1     Running   0           100s
frontend-webapp-revision1-65c49c4776-skrds   1/1     Running   0           100s
payal_kharat@LAPTOP-V1NV0QFT:Que-3$ kubectl get deployment
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
frontend-webapp-revision1          3/3     3             3           104s
-----

payal_kharat@LAPTOP-# cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: frontend-webapp-service
spec:
  type: NodePort
  selector:
    app: frontend-webapp-revision1
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30009
-----

payal_kharat@LAPTOP-# kubectl apply -f service.yaml
service/frontend-webapp-service created
payal_kharat@LAPTOP-# kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP  PORT(S)
AGE
frontend-webapp-service            NodePort       10.96.146.180   <none>
80:30009/TCP                       8s
-----

payal_kharat@LAPTOP-# curl http://localhost:8080
<html>
<body>
  <h1>Host: frontend-webapp-revision1-65c49c4776-skrds</h1>
  Version: 1.1
</body>
</html>
-----

payal_kharat@LAPTOP-# cat configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: frontend-config
  namespace: cloudethix-1
data:
  index.html: |
    <html>
    <body>
    <h1> I am Cloudethix Team, Are you ?!! </h1>
    Version: 1.1
    </body>
    </html>
-----

payal_kharat@LAPTOP-# kubectl apply -f configmap.yaml
configmap/frontend-config created
payal_kharat@LAPTOP-# kubectl get configmap
NAME                DATA   AGE
frontend-config     1       13s
kube-root-ca.crt    1       54m
```

```
-----
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-webapp-revision1
  namespace: cloudehix-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend-webapp-revision1
  template:
    metadata:
      labels:
        app: frontend-webapp-revision1
    spec:
      containers:
        - name: frontend-webapp-revision1
          image: docker.io/payalkharat/cloudehix_configmap_payal:1.0
          ports:
            - containerPort: 80
          volumeMounts:
            - name: frontend-html
              mountPath: /usr/share/nginx/html/
      volumes:
        - name: frontend-html
          configMap:
            name: frontend-config
-----
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/frontend-webapp-revision1 configured
payal_kharat@LAPTOP-# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
frontend-webapp-revision1-d88954b8d-5l22l	1/1	Running	0	17s
frontend-webapp-revision1-d88954b8d-dzx4n	1/1	Running	0	20s
frontend-webapp-revision1-d88954b8d-np25w	1/1	Running	0	19s

```
-----
payal_kharat@LAPTOP-# curl http://localhost:8080
<html>
<body>
<h1> I am Cloudehix Team, Are you ?!! </h1>
Version: 1.1
</body>
</html>
```

=====

Question-4

- Create 1 Public Docker Hub registry named cloudehix_multicontainer_yourname.
- Clone below repository on your system.
<https://github.com/janakiramm/Kubernetes-multi-container-pod.git>
 - Initialize a local repository & copy the code from above repo to your local repository in any of your working branches.
 - Once code is copied , go to the Build directory and build docker image from docker file

and add meaningful tags and push to docker hub repository.

- Now go to the deploy directory and notice the files

Here, web-pod-1.yml file will create the pod with two containers (Multi container). Take a note of labels, name of containers and ports. Also, please make sure you will update the python container image that you have pushed to your docker registry.

- Now, open web-svc.yml file and notice service Type, selectors & targetPort. Apply the file.
- Now open db-pod.yml & notice the labels, name, Image, containerPort and apply the file.
- Now open the db-svc.yml file and notice service Type, selectors & targetPort. Apply the file.
- Once above files are applied, Check that the Pods and Services are created using command line or lens.
- Now, from the command line run below urls & notice the changes.

curl http://*NODE_IP* :*NODE_PORT*/init

Initialize the database with sample schema

- Now it's time to Insert some sample data. Make sure you will use correct *NODE_IP* :*NODE_PORT*

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "1",
"user": "John Doe"}' http://*NODE_IP* :*NODE_PORT*/users/add**

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "2",
"user": "Jane Doe"}' http://*NODE_IP* :*NODE_PORT*/users/add**

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "3",
"user": "Bill Colls"}' http://*NODE_IP* :*NODE_PORT*/users/add**

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "4",
"user": "Mike Taylor"}' http://*NODE_IP* :*NODE_PORT*/users/add**

- Now access the data that we have added to database using below command.

curl http://*NODE_IP* :*NODE_PORT*/users/1

- The second time you access the data, it appends '(c)' indicating that it is pulled from the Redis cache.
- Also, try to access mysql shell i.e db pod & run select * from the users table. check app.py for DB related information.
- Prepare proper documentation in brief & write start to end flow. Refer below link if you face any issues.

<https://github.com/janakiramm/Kubernetes-multi-container-pod-1>

System Output

```
=====
```

```
payal_kharat@LAPTOP-# git clone https://github.com/janakiramm/Kubernetes-  
multi-container-pod.git  
Cloning into 'Kubernetes-multi-container-pod'...  
remote: Enumerating objects: 51, done.  
remote: Total 51 (delta 0), reused 0 (delta 0), pack-reused 51 (from 1)
```

```
Receiving objects: 100% (51/51), 88.14 KiB | 820.00 KiB/s, done.
Resolving deltas: 100% (21/21), done.
```

```
payal_kharat@LAPTOP-# ls
Kubernetes-multi-container-pod
payal_kharat@LAPTOP-# cp -rf Kubernetes-multi-container-pod/* .
payal_kharat@LAPTOP-# ls
Build  Deploy  Kubernetes-multi-container-pod  README.md  multi-container-
pod.png
```

```
-----
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/Assignment-3/Que-4/.git/
payal_kharat@LAPTOP-# git branch -M master
```

```
-----
payal_kharat@LAPTOP-# git status
On branch master
```

```
No commits yet
```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Build/
    Deploy/
    README.md
    multi-container-pod.png
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
payal_kharat@LAPTOP-# git add .
payal_kharat@LAPTOP-# git commit -m " add multicontainer app"
[master (root-commit) 3306377] add multicontainer app
12 files changed, 279 insertions(+)
create mode 100644 Build/Dockerfile
create mode 100644 Build/app.py
create mode 100644 Build/docker-compose.yml
create mode 100644 Build/requirements.txt
create mode 100644 Deploy/db-pod.yml
create mode 100644 Deploy/db-svc.yml
create mode 100644 Deploy/web-pod-1.yml
create mode 100644 Deploy/web-pod-2.yml
create mode 100644 Deploy/web-rc.yml
create mode 100644 Deploy/web-svc.yml
create mode 100644 README.md
create mode 100644 multi-container-pod.png
```

```
-----
payal_kharat@LAPTOP-# cat web-pod-1.yml
apiVersion: "v1"
kind: Pod
metadata:
  name: web1
  namespace: cloudethix-1
  labels:
    name: web
    app: demo
spec:
  containers:
    - name: redis
      image: redis
      ports:
        - containerPort: 6379
          name: redis
          protocol: TCP
    - name: python
      image: payalkharat/cloudethix_multicontainer_payal:python-v1
      env:
        - name: "REDIS_HOST"
          value: "localhost"
      ports:
        - containerPort: 5000
          name: http
          protocol: TCP
-----
```

```
payal_kharat@LAPTOP-# cat web-svc.yml
apiVersion: v1
kind: Service
metadata:
  name: web
  namespace: cloudethix-1
  labels:
    name: web
    app: demo
spec:
  selector:
    name: web
  type: NodePort
  ports:
    - port: 80
      name: http
      targetPort: 5000
      protocol: TCP
-----
```

```
payal_kharat@LAPTOP-# kubectl apply -f web-svc.yml
service/web created
payal_kharat@LAPTOP-# kubectl get svc
NAME      TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
web       NodePort    10.96.171.18    <none>            80:30771/TCP     49s
-----
```

```
payal_kharat@LAPTOP-# cat db-pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: mysql
  namespace: cloudethix-1
  labels:
```

```

  name: mysql
  app: demo
spec:
  containers:
    - name: mysql
      image: mysql:8.0
      ports:
        - containerPort: 3306
          protocol: TCP
      env:
        - name: "MYSQL_ROOT_PASSWORD"
          value: "password"
      resources:
        requests:
          memory: "512Mi"
          cpu: "250m"
        limits:
          memory: "2Gi"
          cpu: "1"

```

```

payal_kharat@LAPTOP-# cat db-svc.yml
apiVersion: v1
kind: Service
metadata:
  name: mysql
  namespace: cloudethix-1
  labels:
    name: mysql
    app: demo
spec:
  ports:
    - port: 3306
      name: mysql
      targetPort: 3306
  selector:
    name: mysql
    app: demo

```

```

payal_kharat@LAPTOP-# kubectl get pods -o wide
NAME      READY   STATUS             RESTARTS   AGE   IP            NODE
mysql     1/1     Running            0           5s    10.244.2.15   mycluster-worker
mycluster-worker <none>      <none>
web1      2/2     Running           12 (110s ago)  46m   10.244.1.14   mycluster-worker2
worker2   <none>      <none>
payal_kharat@LAPTOP-# kubectl get svc
NAME      TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
mysql     ClusterIP   10.96.71.78   <none>       3306/TCP         48m
web       NodePort    10.96.171.18  <none>       80:30771/TCP     54m

```

```

payal_kharat@LAPTOP-# curl http://localhost:8080/init
DB Init done

```

```

payal_kharat@LAPTOP-# curl -i -H "Content-Type: application/json" -X POST -d
'{"uid":"1","user":"John Doe"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15

```

Date: Sun, 06 Sep 2026 14:15:27 GMT

```
payal_kharat@LAPTOP-# curl -i \ $ curl -i \ -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"2","user":"Jane Doe"}' \
http://localhost:8080/users/add
curl: (3) URL rejected: Bad hostname
curl: (6) Could not resolve host: curl
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Sun, 06 Sep 2026 14:15:45 GMT
```

```
payal_kharat@LAPTOP-# curl -i \ -i \ -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"3","user":"Bill Colls"}' \
http://localhost:8080/users/add
curl: (3) URL rejected: Malformed input to a URL function
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Sun, 06 Sep 2026 14:15:53 GMT
```

```
payal_kharat@LAPTOP-# curl -i \ -i \ -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"4","user":"Mike Taylor"}' \
http://localhost:8080/users/add
curl: (3) URL rejected: Malformed input to a URL function
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Sun, 06 Sep 2026 14:16:04 GMT
```

Added

```
payal_kharat@LAPTOP-# curl http://localhost:8080/users/1
John Doe
payal_kharat@LAPTOP-# curl http://localhost:8080/users/2
Jane Doe
payal_kharat@LAPTOP-# curl http://localhost:8080/users/3
Bill Colls
payal_kharat@LAPTOP-# curl http://localhost:8080/users/4
Mike Taylor
```

```
payal_kharat@LAPTOP-# kubectl exec -it mysql -- bash
bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
Server version: 8.0.46 MySQL Community Server - GPL
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> SHOW DATABASES;
+-----+
| Database          |
+-----+
| USERDB            |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+-----+
5 rows in set (0.03 sec)
```

```
mysql> USE USERDB;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

```
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_USERDB |
+-----+
| users             |
+-----+
1 row in set (0.01 sec)
```

```
mysql> SELECT * FROM users;
+-----+-----+
| ID  | USER          |
+-----+-----+
| 1   | John Doe      |
| 2   | Jane Doe      |
| 3   | Bill Colls    |
| 4   | Mike Taylor   |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql>

=====
```

Question-5

Create 1 Public Docker Hub registry named cloudethix_Initcontainer_yourname.

- Clone below repository on your system.
<https://github.com/janakiramm/simpleapp.git>
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.
- Once Images are pushed to Docker hub registries, create a directory named kube.

Inside the kube directory create deployment.yaml file with 3 replication , label app: simpleapp-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.

- Create one service.yaml file with type nodeport & select simpleapp-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
- Open the webpage in the browser and notice the changes and capture the snap.
- Then delete the deployment that you have just created.
- Update the deployment.yaml file and add volumeMounts with mountPath /usr/share/nginx/html from emptyDir: {} volume.

• Once above changes are added, add initContainers block with below parameters. Also add volumeMounts for Init Container with mountPath "/work-dir" from emptyDir: {} volume.

initContainers:

- name: install
image: busybox:1.28
command:
 - wget
 - "-O"
 - "/work-dir/index.html"
 - <http://info.cern.ch>
- volumeMounts:
 - name: workdir
mountPath: "/work-dir"
 - Add volumes with emptyDir: {} in deployment.yaml file.
 - Once the deployment.yaml file is ready, create the deployment & access the page in the browser and notice the changes.
 - Prepare a well formatted document and write your understanding step by step.

System / Terminal Output

```
=====
=

payal_kharat@LAPTOP-# git clone https://github.com/janakiramm/simpleapp.git
Cloning into 'simpleapp'...
remote: Enumerating objects: 47, done.
remote: Total 47 (delta 0), reused 0 (delta 0), pack-reused 47 (from 1)
Receiving objects: 100% (47/47), 8.20 KiB | 311.00 KiB/s, done.
Resolving deltas: 100% (9/9), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-5$ ls
simpleapp
payal_kharat@LAPTOP-V1NV0QFT:Que-5$ cp -rf simpleapp/* .
payal_kharat@LAPTOP-V1NV0QFT:Que-5$ ls
Dockerfile  html  simpleapp  wrapper.sh
-----

payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
```

```
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/Assignment-3/Que-5/.git/
payal_kharat@LAPTOP-V1NV0QFT:Que-5$ git branch -M master
-----

-
payal_kharat@LAPTOP-# cat deployment.yaml service.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simpleapp-webapp
  namespace: cloudethix-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: simpleapp-webapp
  template:
    metadata:
      labels:
        app: simpleapp-webapp
    spec:
      containers:
        - name: simpleapp
          image:
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
          ports:
            - containerPort: 80
apiVersion: v1
kind: Service
metadata:
  name: simpleapp-webapp-service
  namespace: cloudethix-1
spec:
  type: NodePort
  selector:
    app: simpleapp-webapp
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
-----

payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/simpleapp-webapp created
payal_kharat@LAPTOP-# kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
simpleapp-webapp     3/3     3            3           80s
payal_kharat@LAPTOP-# kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
simpleapp-webapp-6564955fc-8dbk5    1/1     Running   0          82s
simpleapp-webapp-6564955fc-nlztg    1/1     Running   0          82s
simpleapp-webapp-6564955fc-snfth    1/1     Running   0          82s
-----
```

```
-
payal_kharat@LAPTOP-# kubectl apply -f service.yaml
service/simpleapp-webapp-service created
payal_kharat@LAPTOP-# kubectl get svc
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)
AGE
simpleapp-webapp-service            NodePort            10.96.159.76    <none>
80:30080/TCP                      5s
-----

payal_kharat@LAPTOP-# kubectl delete deployment simpleapp-webapp
deployment.apps "simpleapp-webapp" deleted from cloudeithx-1 namespace
payal_kharat@LAPTOP-# kubectl get pods
No resources found in cloudeithx-1 namespace.
-----

-
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/simpleapp-webapp created
payal_kharat@LAPTOP-# kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
simpleapp-webapp     3/3     3            3           13s
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
simpleapp-webapp-7748bd6596-kf65r    1/1     Running   0          17s
simpleapp-webapp-7748bd6596-nrr5c    1/1     Running   0          17s
simpleapp-webapp-7748bd6596-xp9kl    1/1     Running   0          18s
-----

-
payal_kharat@LAPTOP-# curl http://localhost:8080/
<html><head></head><body><header>
<title>http://info.cern.ch</title>
</header>

<h1>http://info.cern.ch - home of the first website</h1>
<p>From here you can:</p>
<ul>
<li><a href="http://info.cern.ch/hypertext/WWW/TheProject.html">Browse the
first website</a></li>
<li><a href="http://line-
mode.cern.ch/www/hypertext/WWW/TheProject.html">Browse the first website using
the line-mode browser simulator</a></li>
<li><a href="http://home.web.cern.ch/topics/birth-web">Learn about the birth
of the web</a></li>
<li><a href="http://home.web.cern.ch/about">Learn about CERN, the physics
laboratory where the web was born</a></li>
</ul>
</body></html>

=====
```

Question-6

Create 1 Public Docker Hub registry named cloudeithx_hpa_yourname.

- Clone below repository on your system.

<https://github.com/vivekamin/kubernetes-hpa-example.git>

- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.

- Once code is copied , build a docker image from the docker file and add meaningful tags and push to the docker hub repository.
- Once the image is pushed, go to k8s directory and update deployment.yaml file with image name from your repo. And then create it.
- Open service.yml and change the type to nodePort and apply the same.
- Open the HPA.yaml file, notice it and then apply the same.
- Open the browser, and access the webpage.
- Now it's time to test the HPA working with the below command

System Output

```
=====
=
payal_kharat@LAPTOP-# git clone https://github.com/vivekamin/kubernetes-hpa-
example.git
Cloning into 'kubernetes-hpa-example'...
remote: Enumerating objects: 26, done.
remote: Total 26 (delta 0), reused 0 (delta 0), pack-reused 26 (from 1)
Receiving objects: 100% (26/26), done.
Resolving deltas: 100% (9/9), done.
payal_kharat@LAPTOP-# cp -rf kubernetes-hpa-example/* .
payal_kharat@LAPTOP-# ls
-----

payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/Assignment-3/Que-6/.git/
-----

payal_kharat@LAPTOP-# git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    README.md
    k8s/
    package.json
    src/

nothing added to commit but untracked files present (use "git add" to track)
payal_kharat@LAPTOP-V1NV0QFT:Que-6$ git add .
payal_kharat@LAPTOP-V1NV0QFT:Que-6$ git commit -m "HPA application revision1"
```

```
[master (root-commit) a0e0b1c] HPA application revision1
7 files changed, 145 insertions(+)
create mode 100644 Dockerfile
create mode 100644 README.md
create mode 100644 k8s/deployment.yml
create mode 100644 k8s/hpa.yml
create mode 100644 k8s/service.yml
create mode 100644 package.json
create mode 100644 src/index.js
```

```
-
payal_kharat@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: node-example
  labels:
    app: node-example
spec:
  replicas: 1
  selector:
    matchLabels:
      app: node-example
  template:
    metadata:
      labels:
        app: node-example
    spec:
      containers:
      - name: node-example
        image: payalkharat/cloudethix_hpa_payal:hpa-v1
        imagePullPolicy: Always
        ports:
        - containerPort: 3000
      resources:
        limits:
          cpu: "0.5"
        requests:
          cpu: "0.25"
```

```
payal_kharat@LAPTOP-# cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: node-example
  namespace: cloudethix-1
  labels:
    app: node-example
spec:
  selector:
    app: node-example
  ports:
  - port: 3000
    protocol: TCP
    nodePort: 30001
  type: NodePort
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yml
deployment.apps/node-example created
payal_kharat@LAPTOP-# kubectl get pods
```

```
NAME                                READY   STATUS    RESTARTS   AGE
node-example-65bb46b959-qbr25      1/1     Running   0           2m21s
-----

payal_kharat@LAPTOP-# kubectl apply -f service.yml
service/node-example created
payal_kharat@LAPTOP-# kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
node-example        NodePort    10.96.37.216    <none>        3000:30001/TCP   2s
-----

payal_kharat@LAPTOP-# cat hpa.yml
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
  annotations:
    name: node-example
    namespace: default
spec:
  maxReplicas: 4
  minReplicas: 1
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: node-example
  targetCPUUtilizationPercentage: 1
-----

payal_kharat@LAPTOP-# kubectl get pods -n kube-system | grep metrics
metrics-server-f5cfcfb64-4fgqd      1/1     Running   0
24s
payal_kharat@LAPTOP-# kubectl patch deployment metrics-server -n kube-system -
-type='json' -p='[{"op":"add","path":"/spec/template/spec/containers/0/args/-
","value":"--kubelet-insecure-tls"}]'
```

deployment.apps/metrics-server patched

```
payal_kharat@LAPTOP-# kubectl rollout restart deployment metrics-server -n
kube-system
deployment.apps/metrics-server restarted
-----

payal_kharat@LAPTOP-# kubectl get hpa
NAME                REFERENCE                TARGETS      MINPODS  MAXPODS
REPLICAS  AGE
node-example        Deployment/node-example   cpu: 0%/1%   1         4         1
18s
-----

-
payal_kharat@LAPTOP-# curl http://localhost:8080
Hello World!!
payal_kharat@LAPTOP-# kubectl get hpa
NAME                REFERENCE                TARGETS      MINPODS  MAXPODS
REPLICAS  AGE
node-example        Deployment/node-example   cpu: 0%/1%   1         4         1
4m9s
=====
```

Question-7

Create 1 Public Docker Hub registry named cloudethix_cronjob_yourname.

- Initialize a local repository & copy below code (three files) to your local repository in any of your working branch.
- Once code is copied, build the docker image from Dockerfile , add meaningful tags and then push the docker image to Docker hub registry.
- Now update the pythoncronjob.yml file to change the image name that you have just pushed to docker hub registry.
- Now create a cron job using pythoncronjob.yml file. Check with kubectl command if the cron job is created.
- Check the Job name which is created by cronjob from command line or lens.
- Then check the pod logs which are created by the job and capture the output.
- Prepare well formatted documents and write your understanding.

System / Terminal Output

```
=====

payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/Assignment-3/Que-7/.git/
-----

payal_kharat@LAPTOP-# vim Dockerfile
payal_kharat@LAPTOP-# vim helloworld.py
payal_kharat@LAPTOP-# vim pythoncronjob.yaml
payal_kharat@LAPTOP-# cat pythoncronjob.yaml
apiVersion: batch/v1
kind: CronJob
metadata:
  name: python-helloworld
  namespace: cloudethix-1
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: python-helloworld
              image: payalkharat/cloudethix_cronjob_payal:cronjob-v2
              command: ["/app/helloworld.py"]
          restartPolicy: OnFailure
-----
```

```
payal_kharat@LAPTOP-#$ kubectl apply -f pythoncronjob.yaml
cronjob.batch/python-helloworld created
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
node-example-65bb46b959-m9q8b      1/1     Running   0           9m59s
-----
-
payal_kharat@LAPTOP-# kubectl get cronjob
NAME                SCHEDULE          TIMEZONE    SUSPEND    ACTIVE   LAST SCHEDULE
AGE
python-helloworld  */1 * * * *      <none>      False     0         43s
3m19s
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
node-example-65bb46b959-m9q8b      1/1     Running   0           13m
python-helloworld-29811888-5j6d5    0/1     Completed 0           2m55s
python-helloworld-29811889-9fk8g    0/1     Completed 0           115s
python-helloworld-29811890-wwdf2    0/1     Completed 0           55s
-----
-
payal_kharat@LAPTOP-# kubectl logs python-helloworld-29811889-9fk8g
Welcome to the Cloudethix World
Today is
2026-09-07 05:33:00.852079

=====
```